

TITLE PAGE

- Problem Statement ID –13
- Problem Statement Title-AlzAware: A Multimodal Human-Centered AIML Framework for Early Alzheimer Risk Prediction and Emotional Monitoring
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Objectives

AlzAware: A Multimodal Human-Centered AIML Framework for Early Alzheimer Risk Prediction and Emotional Monitoring

To develop a multimodal ML-based system for early cognitive risk prediction
Analyze behavioral and cognitive indicators to identify early signs of potential cognitive decline.

To collect and analyze real-time behavioral data
Capture and process speech patterns, typing behavior, emotional responses, and cognitive activity continuously for analysis.

To monitor cognitive decline using digital biomarkers
Analyze indicators such as speech hesitation, memory performance, response delay, and typing dynamics to detect abnormal cognitive trends.

To recognize and monitor patient emotions
Detect emotions such as confusion, stress, anxiety, and frustration using facial expression and voice analysis techniques.

To provide caregiver support through smart monitoring
Design a caregiver dashboard that displays cognitive trends, emotional status, and intelligent risk alerts for timely intervention.

To develop an adaptive AI assistant
Create an intelligent assistant that adjusts communication style and support based on the patient's emotional and cognitive condition.

To implement multimodal AI-based decision making
Combine speech, emotion, behavioral, and cognitive features using machine learning and deep learning models for accurate prediction.

Motivation

Why This Project?



Traditional diagnosis methods:

- Are expensive
- Require hospital visits
- Are not continuous
- And cannot monitor emotional wellbeing effectively

Patients also suffer from:

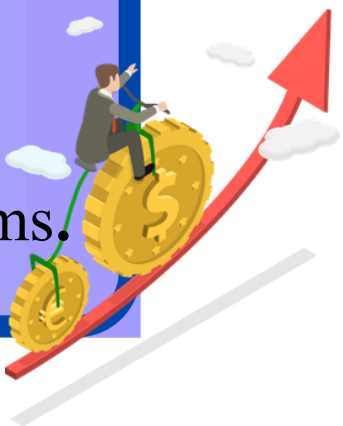
- Anxiety
- Confusion
- Memory loss
- Emotional instability

Existing AI systems mainly focus only on disease classification.

Motivation Behind AlzAware

AlzAware aims to:

- Predict cognitive decline earlier.
- Continuously monitor behavioral patterns.
- Emotionally support patients.
- Reduce caregiver burden.
- Improve quality of life using intelligent AI systems.



Unlike existing Alzheimer's detection systems that rely on clinical biomarkers and static classification models, AlzAware uses multimodal digital behavioral biomarkers, emotion recognition, and continuous monitoring to provide early cognitive risk assessment and caregiver-centered support.

PROPOSED SOLUTION

AlzAware: A Multimodal Human-Centered AI Framework for Early Alzheimer Risk Prediction and Emotional Monitoring

Alzheimer Prediction Using MRI Scans

Analyzes brain MRI scans using deep learning to identify patterns associated with Alzheimer's disease.

Speech & Typing Analysis

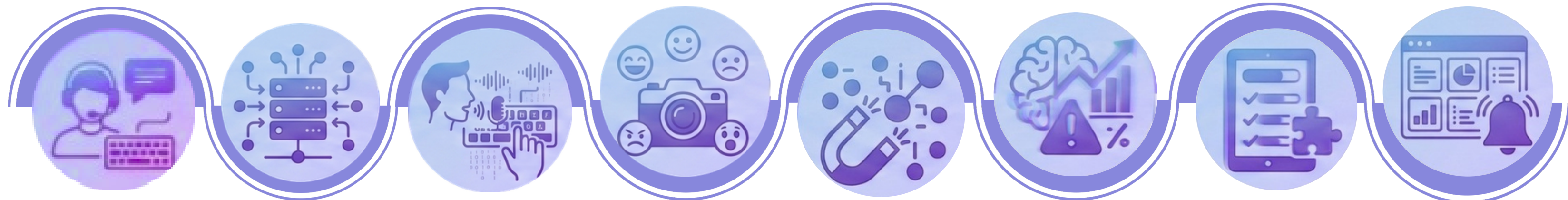
Analyzes pauses, typing speed, hesitation, and repetition patterns.

Feature Extraction

Extracts meaningful cognitive and emotional features from collected data.

Cognitive Assessment

Evaluates memory, reaction time, and recall performance using mini-games.



Real-Time Data Collection

Collects behavioral and emotional data continuously.

Emotion Recognition

Detects stress, confusion, anxiety, and frustration.

ML-Based Risk Prediction

Uses ML models to predict early cognitive decline risk.

User Interaction

Provides reports, cognitive exercises, AI chatbot support, and personalized health recommendations.

Scope

Technical Scope

- Artificial Intelligence
- Machine Learning
- Deep Learning
- Digital Biomarkers
- Emotion Recognition
- Human-Centered Healthcare AI



Application Scope

The system can be used in:

- Hospitals
- Elderly care centers
- Home monitoring systems
- Telemedicine platforms
- Cognitive healthcare research

Future Scope

- Wearable sensor integration
- Mobile app deployment
- Multilingual interaction support
- Real-time doctor connectivity
- Cloud-based healthcare analytics
- Integration with smart healthcare devices

Tech stack & System Workflow

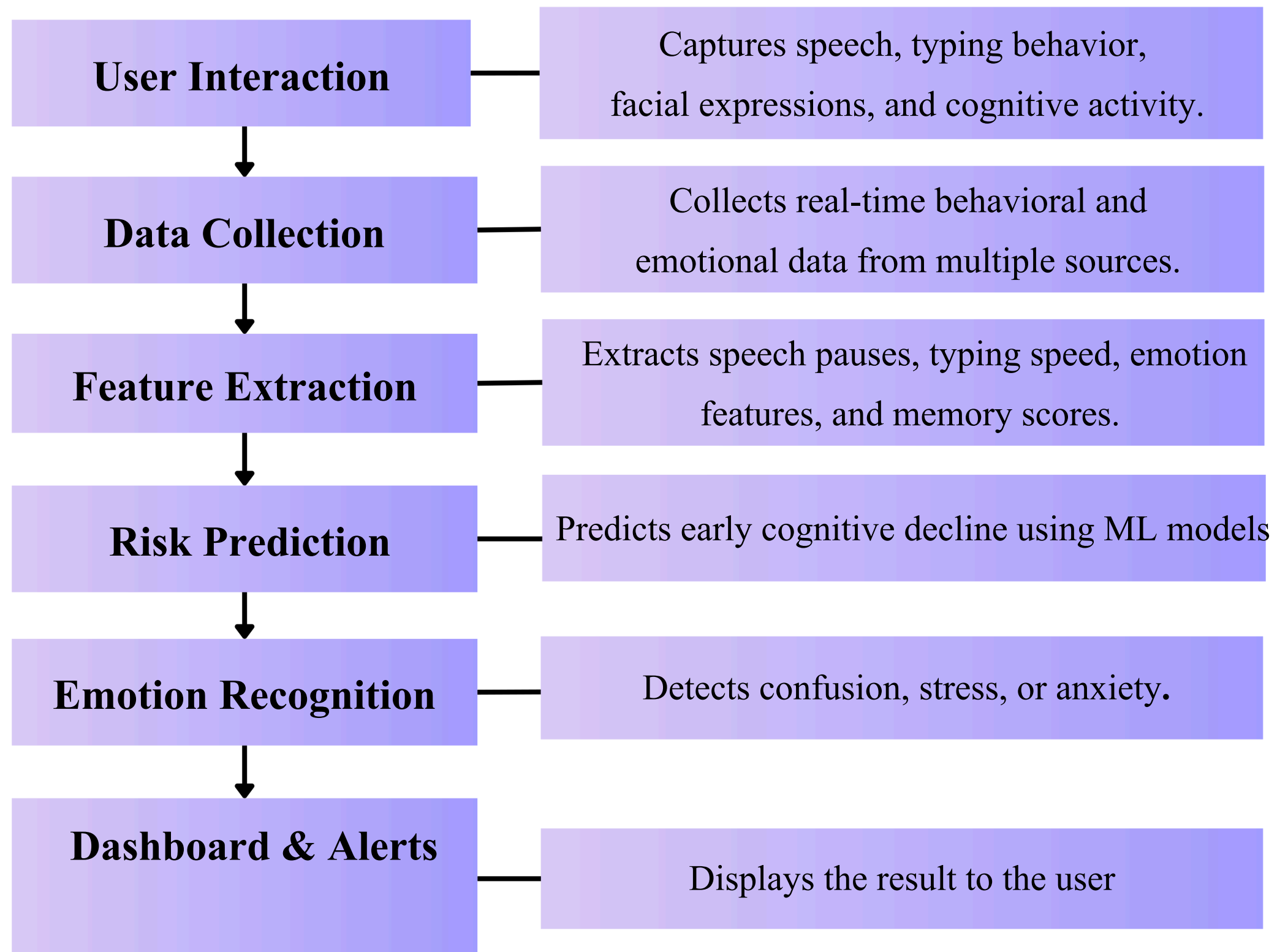
Tech Stack:

Hardware:

- Laptop/Webcam (Facial emotion detection)
- Microphone (Speech analysis)
- Keyboard/Mobile Input (Typing behavior tracking)

Software:

- Python(Core programming)
- TensorFlow / PyTorch (Deep learning models)
- OpenCV (Face detection)
- MediaPipe / DeepFace (Emotion recognition)
- Scikit-learn (Machine learning algorithms)
- Streamlit / Flask (Dashboard & UI)





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Thank-You!